

GANDU THIRUMALESH

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CAREER OBJECTIVE

Graduate in Electrical and Electronics Engineering with 9 months of internship experience in Embedded Systems. Proficient in C, C++, and Python, with hands-on experience in embedded firmware development, Linux-based system programming, hardware interfacing, and peripheral communication protocols (UART, SPI, I2C). Seeking to contribute to complex embedded software architecture and system-level development while continuously learning advanced OS concepts and driver-level programming.

EDUCATIONAL QUALIFICATIONS

B.Tech – Electrical and Electronics Engineering (2021–2024)

Swarna Bharathi Institute of Technology, JNTUH, Khammam

CGPA: 6.72

Diploma – Electrical and Electronics Engineering (2018–2021)

SVS Polytechnic, Bheemaram, Warangal

CGPA: 7.5

SSC (2018)

Resonance Info School, Khammam

CGPA: 8.5

INTERNSHIP EXPERIENCE

Embedded Systems & Hardware Design Intern – Falconx Electro Technologies Pvt.Ltd., Hyderabad

- Developed embedded firmware for STM32, 8051, ESP32, and ATmega328P using C
- Implemented real-time applications using UART, I2C, SPI, ADC, GPIO, PWM, and USB • Interfaced Gas and Temperature sensors, LCD displays, and GSM modules
- Designed a USB-to-UART converter using CH340G in KiCad (schematic & PCB layout)
- Worked with datasheets, schematics, and peripheral protocols
- Explored Embedded Linux concepts including device drivers, IPC, and debugging using GDB, printk and dmesg.

PROJECTS

Industrial Automation Using Embedded Systems (ATmega328P)

- Controlled appliances using HC-05 Bluetooth and GSM modules via AT commands over UART
- Implemented automatic lighting using LDR and fire sensors
- Designed interrupt-based emergency response logic
- Developed modular embedded C firmware

Obstacle-Avoiding Robot Using ATmega328P

- Designed software architecture for an autonomous robot
- Used ultrasonic sensors for obstacle detection and distance measurement
- Developed Embedded C logic for real-time motor control and decision-making
- Interfaced L293D motor driver for bidirectional control

Gas & Temperature Monitoring System (ESP32)

- Integrated MQ Gas Sensor and LM35 Temperature Sensor
- Displayed real-time values on 16×2 LCD

- Implemented threshold-based alert logic using buzzer

USB to UART Converter (KiCad)

- Designed USB-to-Serial converter using CH340G IC
- Integrated 12 MHz crystal oscillator and USB Type-A connector
- Performed ERC/DRC checks and generated 3D PCB model
- Added passive components for signal integrity and voltage regulation

TECHNICAL SKILLS

Programming: C (Advanced), Embedded C, Python (Basic)

Microcontrollers: 8051, Arduino UNO/Nano, ATmega328P, ESP32, STM32

Protocols: ADC, GPIO, I2C, PWM, RS232, SPI, UART, USB

Tools: Keil, CodeVisionAVR, STM32CubeIDE, Proteus

Debugging: GDB, Keil Simulations, Serial Monitors

Engineering Tools: KiCad, AutoCAD Electrical, Cadence, MATLAB Simulink

OS & Systems: Embedded Linux, Linux Boot Process, Multi-threading, IPC

SOFT SKILLS

Analytical thinking, problem-solving, adaptability, initiative-driven mindset, strong communication skills, team collaboration, and time management.

DECLARATION

I hereby declare that all the information provided above is true to the best of my knowledge and belief.